Terms of Reference (ToR) for Institutional Contract

Summary:

UNICEF has been a key partner to the Directorate of Primary Education (DPE) in establishing and scaling IPEMIS as a national platform for data-driven governance. The system has significantly improved the availability, timeliness, and reliability of education data, enabling real-time decision-making, strengthened monitoring, and evidence-based planning. It is widely used across all levels—from central policy-making to school-level functions such as School Planning (SLIP). Teachers and administrators also benefit from improved management of deployment, transfers, and performance data. With evolving sector priorities, there is now a strong focus on enhancing teacher support, accountability, and system efficiency through strengthening teacher data, training systems, Learning Management System (LMS), and revising key modules.

A qualified technical vendor with strong expertise in EMIS development and system integration will be engaged to support continuity, accelerate implementation, and minimize operational risks. Given the integrated and complex nature of IPEMIS, the selected vendor must demonstrate proven capabilities in system enhancement, data interoperability, and managing large-scale education governance platforms. Effective collaboration with government counterparts and full compliance with national standards will be essential for successful delivery.

This assignment will follow a competitive selection process among all qualified UNICEF Long-Term Agreement (LTA) holders within the relevant service category.

Title	Technical Support for Strengthening, and Enhancement of Integrated Primary Education Management Information System (IPEMIS)
LTA No.	Multiple LTAs
Purpose	To recruit a technical agency to support the Directorate of Primary Education (DPE) in completing the government requirements for IPEMIS, including system enhancement, integration, and capacity building.
Location	Dhaka, Bangladesh (with field-level engagement)
Estimated Duration	Total 36 Months Proposed Start Date: 01 August 2026.
Reporting to Technical Supervisor of this assignment	Education Programme

1. Background

The Ministry of Primary and Mass Education (MoPME) and the Directorate of Primary Education (DPE) manage one of the largest primary education systems in the world, serving over 20 million students across more than 120,000 schools. Effective governance of such a large system requires timely, reliable, and integrated data to support planning, monitoring, and decision-making. Historically, the education sector relied on fragmented and manual data systems, resulting in duplication, inconsistencies, and limited use of data for evidence-based decision-making. To address these challenges, UNICEF, in collaboration with DPE and development partners, supported the development of the Integrated Primary Education Management Information System (IPEMIS) under PEDP4.

The development of IPEMIS followed a phased approach. It began with a comprehensive functional and technical assessment of existing EMIS systems, identifying key gaps such as lack of interoperability, weak data validation, and limited reporting capabilities. Based on these findings, IPEMIS was designed as an integrated, web-based platform. The system underwent User Acceptance Testing (UAT) in selected districts and was piloted, where feedback from

teachers and education officials informed system refinement. Following successful piloting, IPEMIS was officially launched nationwide in March 2022 and is now operational across the country, providing real-time data for system management. Since its rollout, IPEMIS has expanded to include multiple modules such as APSS, SLIP, Teacher Transfer Management System (TTMS), E-Monitoring, SEND, Non Government School Registration, scholarship and PEPMIS. These modules are widely used by government officials, schools, and teachers for data management, decentralized planning, monitoring, and service delivery.

With the evolving priorities of the sector programme, there is now a strong focus on teacher support and system efficiency. This requires strategic enhancement of IPEMIS, including:

- Strengthening teacher data systems (deployment, needs-based allocation, retirement projections)
- Enhancing teacher training and a Learning Management System (LMS)
- Enhancing monitoring tools for accountability and performance
- Revising key modules such as Annual Primary School Statistics (APSS), School Level Improvement Plan (SLIP), book demand, and distribution systems

These enhancements are critical to align IPEMIS with the next phase of sector priorities, particularly improving teaching quality and learning outcomes. The government has requested comprehensive technical support to complete these requirements and institutionalize IPEMIS as a fully integrated national system.

This assignment seeks to engage a competent agency with a highly skilled team that demonstrates proven expertise in system architecture, business processes, and the education sector ecosystem, while also having the capacity to build and maintain a strong, collaborative working relationship with the Directorate of Primary Education (DPE) at both central and sub-national levels.

2. Objectives, Purpose and Expected Results

Overall Objective

To strengthen and expand IPEMIS by upgrading priority modules, integrating data across platforms, and enhancing system performance and usability, resulting in improved data quality, real-time reporting, and evidence-based decision-making for DPE users at national, sub-national and school levels.

Specific Objectives

1. Upgrade, operationalized existing modules by improving system performance, user interface, and functionality, resulting in improved system processing time and user satisfaction.
2. Establish API-based integration between IPEMIS and key national systems ensuring real-time data exchange and consistency.
3. Develop and deploy interactive dashboards and automated reporting tools for national, sub-national and school level users, enabling decision-making reports to be generated directly from IPEMIS
4. Build national capacity by providing training for DPE officials and field-level users, establish IPEMIS support structures (including resource teams and helpdesk), and ensure that IMD can independently manage Tier-1 system support.

Expected Results

1. **Fully functional and integrated IPEMIS platform:** IPEMIS operates as a stable, secure, and scalable national system, with all core and upgraded modules fully integrated into a unified platform. The system ensures seamless data flow across modules, reduced system downtime, and supports real-time access for users at national, sub-national, and school levels.

2. **All modules operational and linked with core system:** All modules (e.g., Teacher Management, LMS, SEND, CMS, Citizen Service) are fully deployed, tested, and integrated with the existing IPEMIS architecture. These modules function within a single sign-on environment and support automated workflows, ensuring efficient service delivery and data consistency.

3. **Improved data quality, accessibility, and use:** Data within IPEMIS is accurate, validated, and regularly updated, with built-in validation rules and reduced duplication. Users at all levels can easily access disaggregated, real-time data and dashboards, leading to increased use of data for planning, monitoring, and reporting.

4. **Enhanced service delivery to schools, teachers, and citizens:** IPEMIS enables faster, transparent, and user-friendly service delivery, including teacher transfers, SLIP planning, emergency support (EiE), and citizen services. Schools and teachers benefit from reduced administrative burden and improved response time for key services.

5. **Strengthened capacity of DPE technical teams:** DPE's Information Management Division (IMD) and field-level teams are equipped with the skills, tools, and systems to manage, maintain, and utilize IPEMIS effectively. This includes independent handling of routine system operations (Tier-1 support) and improved capacity in data analysis and system management, ensuring long-term sustainability.

3. Description of Assignment

3.1 Scope of Work

A. Upgradation and enhancement of existing Modules:

Teacher Module: Teacher workforce size & composition

The existing Teacher Module of IPEMIS already serves as a core repository capturing essential teacher information such as name, demographic details, address, date of joining, qualifications, and service records, providing a strong foundation for personnel management. However, to respond to emerging sector priorities and ensure an adequate and equitable teacher workforce, the module requires further enhancement to support dynamic workforce planning. Specifically, it will be expanded to include real-time, disaggregated analytics on teacher availability by subject, grade, and location; gender balance; vacancy mapping; and retirement projections. Integrating these features with enrolment and school-level data will enable automated forecasting, scenario analysis, and evidence-based decision-making. Such enhancements will allow the Directorate of Primary Education (DPE) to proactively identify shortages, optimize recruitment and deployment, and align teacher supply with evolving system needs and national education priorities.

Teacher Transfer and Management System (TTMS): Automation of teacher deployment, equity & distribution

The TTMS within IPEMIS currently provides a structured, application-based workflow for managing teacher postings and transfers. It supports transfer applications for Assistant Teachers (AT) and Head Teachers (HT), including punishment transfers at the intra-upazila level for ATs, and enables basic processing, verification, and approval through digital workflows. This has already contributed to improved transparency and reduced manual handling compared to earlier paper-based systems.

However, to fully address sector priorities on equitable and needs-based teacher deployment, the module requires further enhancement. The upgraded TTMS will expand its functionality to include automated workflows for mutual transfers of both AT and HT across upazila, district, and city corporation levels, as well as coordination transfers and attachment transfers, which are currently not covered. In addition, the system will incorporate configurable, policy-aligned business rules (e.g., tenure, subject requirements, hardship criteria, and vacancy data) and enable real-time application tracking, automated notifications, and end-to-end digital approval processes. Integration with teacher profiles, APSS data, and enrolment projections will allow advanced analytics, scenario simulation, and identification of teacher shortages and surpluses. These enhancements will significantly reduce administrative processing time, improve transparency and accountability, and ensure that teacher deployment is aligned with actual school-level needs and national teachers' framework.

Learning Management System (LMS): Teacher Capability, Professional Development

The LMS within IPEMIS already provides a foundational platform for digital learning, enabling the delivery of training content to teachers and supporting basic access to e-learning materials. Developed under earlier requirements, the current system facilitates content sharing and user access; however, it has limitations in ensuring structured learning and effective tracking of teacher development. These include the absence of clearly defined course pathways, limited controls to ensure content completion before assessments, inadequate learner progress tracking, weak assessment validation, and usability constraints such as limited video display controls and lack of multi-tab functionality for integrated learning resources.

To address sector priorities on strengthening teacher capability and continuous professional development, the LMS requires comprehensive enhancement and realignment with current operational and policy needs. The upgraded system will support structured, curriculum-aligned course pathways linked to national teacher competency frameworks, enforce learning prerequisites, and strengthen assessment integrity through controlled access and validation mechanisms. It should enable robust tracking of learner progress, performance, and certification, integrated with teacher profiles for informed decisions on deployment, promotion, and training needs. Enhanced user experience features—including improved multimedia functionality, mobile and offline access, and interactive learning environments—will ensure accessibility and engagement at scale. In addition, integration with a strengthened Content Management System (CMS), real-time dashboards, and feedback mechanisms will transform the LMS into a comprehensive, outcome-driven professional development platform, ensuring that teacher training is systematic, measurable, and directly linked to improved classroom practices and learning outcomes.

Training Tracking System: Pedagogical Practices (Quality of Teaching)

The Training Tracking Module within IPEMIS currently serves as a platform for managing training programmes and maintaining records of teacher and officer participation, including basic functionalities for training registration and attendance tracking. While this provides an important foundation for monitoring training delivery, the module has several functional limitations that constrain its ability to support system-wide professional development and evidence-based decision-making. Notably, it lacks integration with the Learning Management System (LMS) to track e-learning completion, does not adequately differentiate between online and face-to-face training modalities, and has limited capacity for training-wise budget management and dynamic analytics. In addition, the absence of drill-down dashboards and real-time reporting restricts the ability of decision-makers to assess training coverage, equity, and effectiveness across districts and upazilas.

To address sector priorities on strengthening teacher capability and improving pedagogical practices, the module requires comprehensive enhancement. The upgraded system should enable full integration with the LMS and Content Management System (CMS) to create a unified ecosystem where training delivery, content access, and learning outcomes are interconnected. It should support automated certification, enforce participation and completion tracking, and provide differentiated workflows for various training modalities. Advanced analytics, including dynamic dashboards with drill-down features, should be incorporated to monitor training distribution, budget utilization, and impact at all administrative levels. Furthermore, linking training data with E-Monitoring and classroom observation systems will allow tracking of how training translates into improved teaching practices, helping identify gaps and inform targeted interventions. These enhancements will transform the module from a basic administrative tool into a strategic platform for managing continuous professional development, ensuring that teacher training is structured, data-driven, and directly aligned with improved teaching quality and student learning outcomes.

E-Monitoring: Accountability for Learning Outcomes

The E-Monitoring module within IPEMIS currently provides a mobile-based platform for monitoring school visits, enabling inspectors and education officials to capture key information on school functioning, attendance, and basic classroom practices during field visits. This has established an important foundation for strengthening supervision and data collection at the school level. However, the current version has several limitations that restrict its effectiveness in driving accountability for learning outcomes. It does not support configurable inspection schedules for inspectors, dynamic indicator-based monitoring aligned with evolving sector priorities, or an updatable inspection calendar. Additionally, there are gaps in structured follow-up mechanisms, real-time aggregation of data, and actionable feedback loops.

To address sector priorities on accountability and data-driven decision-making, the module requires comprehensive enhancement. The upgraded E-Monitoring system will support configurable, target-based inspection scheduling, flexible calendar management, and dynamic monitoring indicators aligned with national priorities such as foundational learning, inclusive education, and school quality standards. It should incorporate real-time dashboards with aggregated analytics, dynamic scoring systems, and automated feedback and follow-up tracking to ensure that identified issues are addressed systematically. Strengthening data validation, integrating with other IPEMIS modules (such as LMS, APSS, and teacher data), and aligning data collection tools with updated monitoring frameworks will enable more targeted interventions and improve field supervision. These enhancements will transform E-Monitoring from a data collection tool into a robust accountability mechanism that links supervision with performance improvement, ensuring that decisions are evidence-based and focused on improving teaching quality and student learning outcomes.

Annual Primary School Statistics (APSS): Evidence-based planning and accountability

The APSS module within IPEMIS currently provides a structured mechanism for collecting annual school-level data through standardized forms, enabling the Directorate of Primary Education (DPE) to compile essential information on enrolment, teachers, and school conditions. This has established a strong foundation for national education statistics and routine reporting. However, the current system has limitations in supporting advanced data use: the reporting interface lacks dynamic dashboards, comparative analytics, and drill-down functionality, while validation features are limited and not fully aligned with evolving sector indicators and data quality requirements.

To address sector priorities on evidence-based planning and accountability, the module requires significant enhancement. The upgraded APSS will incorporate configurable and session-based data collection forms, allowing flexibility to update indicators as needed, along with automated cross-field validation rules to improve data accuracy and reduce duplication. It should enable role-based workflows for data entry, review, and approval, and integrate with other IPEMIS modules (teacher, student, infrastructure) to allow auto-population of data and minimize manual input. In addition, the system will introduce dynamic dashboards with decision-making parameters, real-time disaggregated reporting, and advanced comparative analytics with drill-down capabilities to support analysis across districts, upazilas, and schools. Historical trend analysis and projection features will further strengthen planning for enrolment, teacher requirements, and resource allocation. These enhancements will transform APSS from a data collection tool into a powerful analytics platform, enabling timely, accurate, and actionable insights to inform policy, planning, and system performance improvement.

Book Distribution Module

The Book Distribution module currently manages textbook demand collection and distribution from national to school level; however, it lacks accurate demand visibility across administrative tiers and does not generate alerts for anomalies. The upgraded module will introduce automated demand generation, role-based demand adjustments, inclusion of non-listed private schools, and GPS mapped geo-location-based tracking. It will also provide real-time distribution monitoring and dynamic dashboards with drill-down features, improving efficiency, reducing delays, and strengthening accountability across the supply chain.

Personnel Data Management (PDM) Module

The PDM module currently maintains profiles of DPE officers and staff but has limited functionality for ACR, leave, and transfer management. The enhanced module will introduce end-to-end ACR workflows, leave and transfer management for both officers and staff, and a dynamic organogram with post adjustments. Advanced reporting and dashboard features will establish a single, reliable source of human resource data for improved workforce planning and management.

School Level Improvement Plan (SLIP) Module

The SLIP module supports school planning and fund management but currently faces challenges such as limitations in multi-year planning adjustments, tracking progress, and flexible budget allocation. The upgraded system will enable year-wise planning and progress tracking, configurable budget distribution, and dynamic adjustment of allocated funds. These enhancements will strengthen transparency, compliance with approval workflows, and financial governance, improving school-level planning effectiveness.

Citizen Service Portal and accountability for affected populations
Currently, citizen services such as requests for documents, teacher information, and feedback are largely manual, leading to delays and lack of tracking. The upgraded Citizen Service Portal will provide a unified digital platform for submitting requests, real-time status tracking, automated notifications, and monitoring dashboards. This will enhance transparency, responsiveness, and citizen engagement in line with the Digital Bangladesh and Smart Bangladesh vision.

Equipment Management System (EQMS)
The EQMS module currently manages ICT equipment distribution and maintenance but does not support inventory management of consumable or operational supplies. The upgraded module will expand to include store inventory management, requisition creation, multi-level approval workflows, and real-time inventory updates. Dynamic monitoring reports will enable better tracking and reduce manual processes, improving overall resource management.

Dynamic Reporting, Dashboards & Analytics
The current system provides predefined reports but lacks a flexible, unified reporting and analytics capability. The enhancement will introduce a dynamic reporting engine, allowing users to generate custom, real-time reports without IT support. It will include interactive dashboards with drill-down capabilities and cross-module analytics, enabling comprehensive data analysis from national to school level and significantly strengthening data-driven decision-making.

B. System Strengthening

Improve UI/UX and dashboards
Redesign the IPEMIS user interface to make it more intuitive, user-friendly, and accessible for different categories of users (national, district, upazila, and school levels). This includes simplifying navigation, reducing the number of steps required to complete tasks, and ensuring compatibility across devices (desktop, tablet, and mobile). Develop role-based, interactive dashboards that present key indicators (e.g., enrollment, teacher deployment, SLIP status, infrastructure needs) in a clear and visual format using charts, maps, and summaries. Ensure dashboards support real-time data visualization and can be customized based on user needs to improve data use for decision-making. Accessibility for person with disability should be ensured in the system.

Enhance system performance and scalability
Strengthen the system architecture to ensure fast response time, high availability, and reliability (page load under 0.3 ms or 99.5% uptime), even with large numbers of concurrent users. Optimize database queries, improve server capacity, and implement load balancing to support nationwide usage. Ensure the system is scalable to accommodate future expansion, including additional modules, increased data volume, and growing user base. Introduce system monitoring tools to track performance, uptime, and errors, enabling proactive maintenance and minimizing downtime.

Implement role-based access control and audit systems
Establish a robust role-based access control (RBAC) framework to ensure that users can only access data and functionalities relevant to their roles and responsibilities. Define clear user roles (e.g., headteacher, district officer, national administrator) with appropriate permissions. Implement audit trails and logging mechanisms to track user activities, data changes, and system access, ensuring transparency and accountability. This will enhance data security, prevent unauthorized access, and support compliance with government data protection standards.

C. Integration & Interoperability
Integration and interoperability will be strengthened to ensure IPEMIS functions as a fully connected and unified data platform. The assignment will include the development of standardized, secure APIs to enable seamless data exchange between IPEMIS and key national systems such as NID, BANBEIS, CRVS, and other relevant databases. This will facilitate real-time synchronization of student, teacher, and school data, reducing duplication and manual entry. The system will also establish data harmonization protocols, including common data standards, validation rules, and mapping mechanisms to ensure consistency, accuracy, and compatibility across systems. These enhancements will

improve efficiency, strengthen data reliability, and support integrated, evidence-based decision-making across the education sector.

D. Capacity Building

Capacity building will be a core component of the assignment to ensure effective use and long-term sustainability of IPEMIS. The technical agency will design and deliver structured training programmes for 2800 DPE officials at national, district, upazila, and school levels, focusing on system operation, data entry, reporting, and use of dashboards for decision-making. Comprehensive user manuals, administrative guides, and technical documentation will be developed to support day-to-day system use and maintenance. In addition, the assignment will support institutionalization of IPEMIS by strengthening the capacity of the Information Management Division (IMD), establishing standard operating procedures, and promoting a system of continuous learning and support (including helpdesk and resource teams), ensuring that DPE can independently manage and sustain the system over time.

4. Deliverables

Tasks/Milestone	Deliverables:	Timeline
Inception phase	Inception report including refined methodology, implementation approach, assessment tools, stakeholder consultation plan, risk mitigation strategy, and detailed workplan	Month 1
Develop structural layout of the intended integrated information system incorporating new modules.	Software Requirement Specification (SRS) Report including system architecture, interoperability framework, module integration roadmap, database design, security framework, scalability requirements, UI/UX improvement strategy, dashboard enhancement plan, and technical specifications for integration of new modules	Month 2
Design/redesigning and upgrading of the modules.	Development, redesign, enhancement, integration, testing, and deployment of the following modules and systems: • Teachers' Module • Learning Management & Training Tracker System • E-Monitoring System • Special Education Needs and Disability (SEND) Module • Annual Primary School Statistics (APSS) Module • Book Demand and Distribution Module • Personnel Data Management) Module • School Level Improvement Plan (SLIP) Management Module • Teacher Transfer System • Citizen Service Module • Equipment Management System (EQMS) Module • Dynamic Reporting and Analytics Module • Education In Emergency (EiE) Module **Other Deliverables** • Improved UI/UX design across web and mobile platforms • Audit Module	Month 02-24

Tasks/Milestone	Deliverables:	Timeline
	• Enhanced dashboards with dynamic analytics and drill-down reporting • Improved system performance, optimization, and scalability • API development and interoperability with external systems/databases • Implementation of role-based access control (RBAC) and audit trail systems • User Acceptance Testing (UAT) reports • System integration and deployment reports • Technical documentation and source code repository updates	
Maintenance support to the system, capacity building of DPE officials and teachers.	Comprehensive maintenance and technical support report including bug fixing, system optimization, troubleshooting, and continuous enhancement support. Capacity-building and institutionalization deliverables including: • Training completion reports for DPE officials, teachers, and administrators • User manuals, operational guidelines, and technical documentation • System administration and maintenance training materials • Integration completion and operationalization report • Capacity-building report on system administration, data management, analytics, and reporting • Sustainability and institutionalization roadmap	Month 24-36

5. Reporting requirements

- Inception Report within 2 weeks of signing the contract.
- Progress Reports by 1st of every month.
- Minutes of Meetings within 2 days after each meetings.
- Mission Reports within 10 days of each mission.
- Technical Reports on a monthly/quarterly/six-monthly basis.
- Draft Final Report within 3 weeks of completion of field work.
- Final Report within 2 weeks of receipt of comments by UNICEF technical supervisor.
- Training/presentation material before 2 weeks of the date of the training/presentation.
- Workshop/training evaluation report within 2 weeks of completion of the workshop.

All reports to be submitted in:

- Word (editable)
- PDF (final version)